

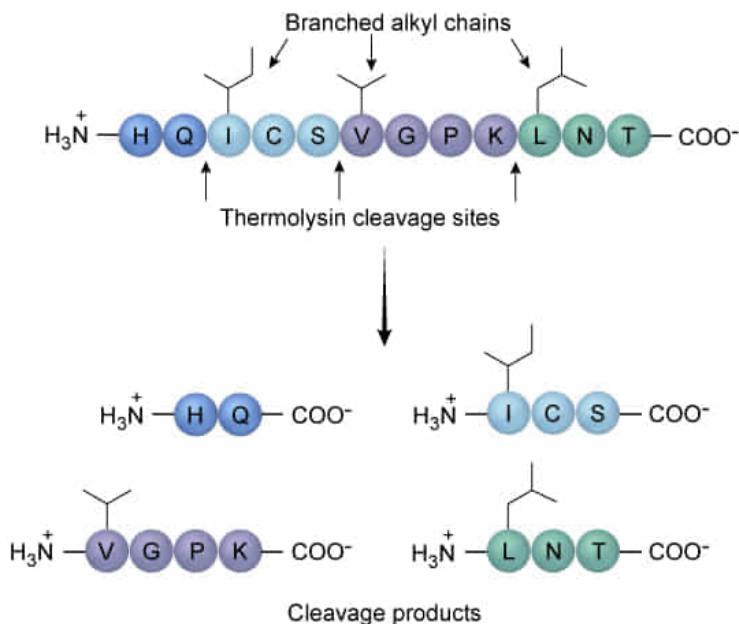
Thermolysin is a protease that can cleave peptides at the N-terminal side of branched alkyl residues. Which of the following peptides has the most thermolysin cleavage sites?

- ✔ ☒ A. HQICSVGPKLNT [60%]
☐ B. DNKGSTVYQRWC [11%]
☐ C. GHPTLDQIW CER [15%]
☐ D. RTHCQGYNESDK [13%]

Correct

60% answered correctly

Explanation:



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The amino acid residues in a peptide can be classified by the general properties of their side chains. Proteases cleave peptide bonds at specific sites by recognizing **residues with specific properties**. They may cleave the bond on the N-terminal side of the residue (the side that is closer to the N-terminus of the peptide) or they may cleave the bond on the C-

terminal side of the recognized residue (the side that is closer to the C-terminus). For example, trypsin cleaves bonds on the C-terminal side of positively charged residues (R, K), and chymotrypsin cleaves bonds on the C-terminal side of large aromatic residues (F, Y, W).

The question states that thermolysin can cleave peptide bonds on the N-terminal side of branched alkyl residues. An alkyl chain is a chemical group that contains only carbon and hydrogen atoms that are not part of an aromatic ring. The amino acids with alkyl side chain groups include [alanine \(A\)](#), [proline \(P\)](#), [leucine \(L\)](#), [isoleucine \(I\)](#), and [valine \(V\)](#). Of these, **only leucine, isoleucine, and valine are branched** (ie, they contain [branch points](#) in their side chains). Therefore, the peptide with the most L, I, and V residues will have the most thermolysin cleavage sites. The peptide **HQICSVGPKLNT** has **three thermolysin cleavage sites**—one each of L, I, and V, which is more than any of the other choices given.

(Choice B) The peptide DNKGSTVYQRWC has only one thermolysin cleavage site at valine.

(Choice C) The peptide GHPTLDQIW CER has only two thermolysin cleavage sites, one at leucine and one at isoleucine.

(Choice D) The peptide RTHCQGYNESDK does not have any L, I, or V residues, and it has no thermolysin cleavage sites.

Educational objective:

Amino acids have side chains with specific properties. Side chains with similar properties can be recognized by the same enzyme. Leucine, isoleucine, and valine are hydrophobic

amino acids with branched alkyl chains.

Time spent:0 sec

Subject:
Biochemistry

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System:
Amino Acids and Proteins